Mississippi Valley State University

# CS 341 — Discrete Structures

## Practice Set 1

*Statements, Truth Values, and Open Sentences*

Covers Chartrand & Zhang, Chapter 0 and Section 1.1

**Status of this set**

This set is **ungraded practice**. It is not submitted for a grade, and the approved solutions are released with it so you can check your own work. Your first *graded* problem set covers Sections 1.1–1.2 and will be posted on Canvas.
Work the problems on your own paper — there is deliberately no space to write here, so that you can attempt a problem more than once. Practice only works if you attempt every problem **before** reading any solution; see "The one rule that matters" below.

# Part I: How to Study for This Course

### This course grades your reasoning, not your answer

In CS 204 a program either produced the right output or it did not. Here, a correct final answer with no reasoning earns very little credit, and a wrong final answer with sound, clearly written reasoning earns a great deal. The written argument *is* the work.

Get used to this now, in Week 1, while the content is easy. The habits are the hard part, not the mathematics.

### The one rule that matters

**Attempt first, then check**

Work every problem completely before you look at the solution — including the ones you are stuck on. Write down where you got stuck and *why*.
Reading a solution you have not attempted feels productive and is not. You will recognize every step, conclude that you understood it, and then be unable to reproduce any of it on a quiz. Recognition is not the same as recall, and only one of the two is graded.

### A workable weekly cycle

1. **Before class:** read the assigned pages with a pen, working each of the book's examples yourself before reading its solution. Expect roughly **ten minutes per page**, not one — if you are reading a mathematics text at the speed you read an article, you are not reading it.

2. **During class:** fill in the guided notes. The parts you write are the parts you retain.
3. **Within 24 hours:** do the practice problems. Memory decays fastest on day one.
4. **Then, and only then:** check against the approved solutions. Mark each problem correct, partially correct, or wrong — and for anything not fully correct, write one sentence naming what you missed.
5. **Office hours:** bring the specific problems you marked, with your attempt on paper. "I don't get it" is hard to help with; "I got stuck at this line, because I wasn't sure whether this counts as a statement" takes about ninety seconds to resolve.

## What a complete written answer looks like

Consider: *Is "$x + 2 = 7$" a statement? Explain.*

| | |
|---|---|
| **No credit** | "No." |
| **Partial** | "No, it's an open sentence." |
| **Full** | "No. It contains the variable $x$, and its truth value depends on the value of $x$: if $x = 5$ it is true, and if $x = 3$ it is false. Because it does not have one definite truth value on its own, it is an open sentence rather than a statement. It becomes a statement once $x$ is assigned a value." |

The full answer names the relevant feature (the variable), shows the consequence with concrete cases, applies the definition, and states the conclusion. Every graded answer in this course should do those four things.

## Collaboration and generative AI, in practice

You may discuss *approaches* with classmates. You must write your final solution yourself, in your own words, from your own understanding.

For graded work, generative AI is limited to grammar, formatting, and LaTeX help — not generating, structuring, or substantially revising your answers. The reason is specific to this course: the skill being assessed is *producing the reasoning*, so outsourcing it removes the only thing the grade measures. You may be asked to explain any submitted step, and being unable to do so is treated as not having done the work.

**Check this before you hand in any written work**

- Did I answer the question that was actually asked?
- Is every claim justified, or did I call something obvious to avoid explaining it?
- Could a classmate who missed class follow my reasoning without asking me anything?
- Did I apply the *definition*, or did I appeal to intuition?
- Are these complete sentences? Mathematics is written in prose, not in fragments.
- Did I state my conclusion explicitly, rather than leaving it to be inferred?

**Five mistakes almost everyone makes in Week 1**

1. **"Statement" means "something I can check."** It does not. A sentence is a statement because it *has* a definite truth value, not because you know that value.
2. **Confusing "unknown" with "neither true nor false."** An unsolved conjecture is still a statement.
3. **Calling an open sentence false.** "$x + 2 = 7$" is not false; it has no truth value at all until $x$ is fixed.
4. **Giving the answer with no reasoning.** See the table above.
5. **Reading "or" as exclusive.** In mathematics "$P$ or $Q$" is true when both hold.

# Part II: Practice Problems

## Section A — Is it a statement?

Classify each sentence as exactly one of the following, and give a one-sentence reason.

| | |
|---|---|
| **true statement** | definitely true |
| **false statement** | definitely false |
| **statement, truth value unknown** | definitely one or the other, but nobody currently knows which |
| **open sentence** | contains a variable; no truth value until the variable is fixed |
| **not a statement** | not the kind of sentence that can be true or false at all |

**1.** 17 is a prime number.

**2.** Is 51 a prime number?

**3.** 51 is a prime number.

**4.** $n$ is a prime number.

**5.** Please submit your homework through Canvas.

**6.** $\sqrt{2}$ is a rational number.

**7.** There is intelligent life on a planet outside our solar system.

**8.** This sentence is false.

**9.** $x^2 \geq 0$, where $x$ is a real number.

**10.** Discrete mathematics is more useful than calculus.

**11.** The $10^{100}$-th decimal digit of $\pi$ is 7.

**12.** Every even integer greater than 2 is the sum of two prime numbers.

## Section B — Open sentences

**13.** Let $P(x) : x^2 - 5x + 6 = 0$, where $x$ is an integer.

(a) Is $P(2)$ true or false?

(b) Is $P(4)$ true or false?

(c) Find *all* integers $x$ for which $P(x)$ is true.

**14.** Let $Q(n) : n^2 + n$ is even, where $n$ is an integer.

(a) Evaluate $Q(3)$, $Q(4)$, and $Q(-5)$.

(b) Based on your results, what do you suspect about $Q(n)$ in general?

(c) Give a reason your suspicion is correct. (You do not need a formal proof yet.)

**15.** Let $R(x) : 2^x > x^2$, where $x$ is a positive integer.

(a) Evaluate $R(x)$ for $x = 1, 2, 3, 4, 5, 6$.

(b) A classmate checks $R(1)$, sees that it is true, and concludes that $R(x)$ is true for every positive integer. What is wrong with that reasoning?

(c) For which positive integers does $R(x)$ appear to be true?

**16.** Let $S(x, y) : x + y = 10$, where $x$ and $y$ are positive integers. How many pairs $(x, y)$ make $S(x, y)$ true? List them.

## Section C — Counting distinct possibilities

**17.** A PIN is exactly four digits, each from 0 through 9, and digits may repeat. How many PINs are possible?

**18.** Same as above, except no digit may be repeated. How many PINs are possible?

**19.** How many binary strings have length exactly 10?

**20.** A password is exactly 8 characters, each chosen from the 26 lowercase letters, 26 uppercase letters, and 10 digits.

(a) How many passwords are possible?

(b) Roughly how many times larger is this than the number of four-digit PINs?

**21.** Classify each quantity as discrete or continuous, and give a brief reason for any two of them.

(a) The number of edges in a network.

(b) The voltage on a wire.

(c) The number of comparisons a sorting algorithm performs.

(d) The temperature of a CPU.

(e) The set of valid IPv4 addresses.

## Section D — Precision in ordinary language and in code

**22.** A course policy reads: *An assignment receives credit if it is submitted before the deadline and passes the automated tests.* For each case, does the assignment receive credit?

(a) Submitted on time; passes the tests.

(b) Submitted on time; fails the tests.

(c) Submitted late; passes the tests.

(d) Submitted late; fails the tests.

**23.** A security requirement reads: *A password is accepted only if it is longer than eight characters and contains at least one symbol.* A student writes:

```
if len(password) > 8 or has_symbol(password):
    accept()
```

(a) Give a specific password this code wrongly accepts.

(b) Correct the code.

**24.** For each sentence, say whether "or" is being used inclusively (both may hold) or exclusively (exactly one holds), and explain how you can tell.

(a) "Bring a pen or a pencil to the exam."

(b) "Your entrée comes with soup or salad."

(c) "To enroll, a student must have completed CS 204 or received instructor permission."

## Section E — Writing

**25.** In four to six sentences, explain to a classmate why "There are 25 students in this room" is a statement even before anyone counts, while "$x + 2 = 7$" is not a statement at all. Your answer should make clear that these two sentences fail to have a known truth value for *different reasons*.

## Section F — Challenge (optional)

**26.** Consider the sentence "This sentence has five words."

(a) Is it a statement? If so, what is its truth value?

(b) Compare it with problem #8. Both sentences refer to themselves. Why is only one of them a problem?

**27.** Suppose nobody in our classroom is over 200 years old. Consider: "Every student in this room who is over 200 years old has a Ph.D." Is this true, false, or neither? Defend your answer. (This one is genuinely contested on first encounter — we settle it formally in Section 1.3.)